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DEPARTMENT OF INFORMATION TECHNOLOGY

IV B. Tech. SEM – I
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PROJECT STAGE-1(PS-1)

Branch/Section: IT - B

Team Number: 13

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Domain Bucket/Name: Artificial Intelligence

Title/Statement: Object Detection using YOLO

Abstract:

Object detection is a fundamental task in computer vision, enabling the identification and localization of objects within an image or video. Among the many techniques developed, the "You Only Look Once" (YOLO) framework stands out due to its efficiency and real-time processing capabilities. YOLO reformulates object detection as a single regression problem, directly predicting bounding boxes and class probabilities from entire images in one evaluation. This approach eliminates the need for region proposal networks and significantly reduces computation time compared to traditional methods like R-CNN and its variants.

This work provides an overview of YOLO's architecture, which includes its grid-based detection mechanism, unified loss function, and end-to-end training methodology. Additionally, it highlights advancements in subsequent versions, such as YOLOv3, YOLOv4, and YOLOv5, focusing on improvements in accuracy, robustness, and scalability. The application of YOLO in diverse fields—autonomous driving, surveillance, medical imaging, and robotics—is discussed to emphasize its versatility. Despite its advantages, YOLO faces challenges, including difficulties with small object detection and complex overlapping scenarios. Strategies to address these issues, such as improved anchor box design and leveraging attention mechanisms, are explored. This study underscores YOLO's contribution to advancing object detection and its potential for further innovation.

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